2024-Winter Paper 1 – Basic Sciences in Urology

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Q1. Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.

Type: LAQ (25 marks) (related: 2019-Summer Paper 1, Q2; 2020-Summer Paper 1, Q3b)

Anatomy: definition and boundaries

- The space behind the parietal peritoneum. **Anterior:** posterior peritoneal reflection; **posterior:** posterior abdominal wall; **cranial:** diaphragm; **caudal:** extraperitoneal pelvis.

Group	Contents
Primarily retroperitoneal (formed there)	Kidneys, ureters, adrenals, aorta, IVC (inferior vena cava)
Secondarily retroperitoneal (mesentery fused to the wall)	Ascending and descending colon, duodenum D2–D3, pancreas (not the tail)
Vessels, lymphatics, nerves	Lumbar nodes, cisterna chyli, sympathetic trunks, autonomic plexuses, lumbosacral plexus

Posterior abdominal wall

- **Flank layers:** external oblique, internal oblique, transversus abdominis, transversalis fascia. **Posterior:** psoas, quadratus lumborum, erector spinae.
- **Lumbodorsal (thoracolumbar) fascia:** three layers that fuse into one at the **tip of the 12th rib**, the standard entry point for retroperitoneoscopy.
- **Ribs:** access above the 12th rib risks the pleura; the intercostal bundle runs under the rib above.

Fasciae and spaces (three embryological strata)

Space	Boundaries	Contents
Anterior pararenal	Posterior peritoneum and anterior renal fascia	Colon, duodenum, pancreas
Perirenal	Anterior (Toldt) and posterior (Zuckerkandl) laminae of Gerota's fascia	Kidney, adrenal, perirenal fat, hilum, upper ureter, gonadal vessels

Space	Boundaries	Contents
Posterior pararenal	Posterior lamina and transversalis fascia	Fat only

- The laminae fuse laterally as the **lateroconal fascia**. The perirenal cone is **open inferiorly**, so urine, blood or pus track into the pelvis.
- Perirenal fat is fine and pale; pararenal fat is coarse and yellow-orange.

Vessels, lymphatics and nerves

- **Aorta** enters at T12. Unpaired anterior branches supply the gut (coeliac, SMA (superior mesenteric artery), IMA (inferior mesenteric artery)); paired lateral branches supply adrenals, kidneys (L1–L2) and gonads; lumbar arteries go to the wall.
- At the hilum the **renal vein lies anterior to the artery**. The **left renal vein is long** (gonadal, adrenal and lumbar tributaries); the right is short.
- Lymph flows right to left; **cisterna chyli** at L1–L2.
- **Nerves at risk:** sympathetic trunk (medial to psoas), genitofemoral (on psoas), subcostal, iliohypogastric, ilioinguinal (on quadratus); hypogastric plexus (ejaculation).

Principles of laparoscopic retroperitoneal surgery

1. **No natural cavity:** the working space must be created (finger and balloon dissection) before insufflation.
2. **Psoas is the key landmark:** it forms the floor. Keep it horizontal on screen; the kidney in Gerota's fascia lies anterior, and the peritoneum is pushed medially.
3. **Direct posterior route to the hilum:** the renal artery, posterior to the vein, is met first, allowing early arterial control.
4. **Peritoneum stays intact:** no bowel mobilisation or adhesiolysis; no spill of tumour, urine or pus into the peritoneum.
5. **Small space:** closely spaced ports, limited triangulation, lens smudging.
6. **Gas behaviour:** CO_2 (carbon dioxide) absorbs through fat, so hypercarbia and subcutaneous emphysema are commoner; haemodynamic swings are smaller than transperitoneal.

Preferred for	Limitations
Adhesions, peritoneal dialysis, morbid obesity, pregnancy; posterior tumours, pyeloplasty, cyst surgery, renal biopsy	Small space, unfamiliar landmarks, harder suturing, learning curve

- **Evidence:** randomised trials and meta-analysis: blood loss, complications and oncology similar to transperitoneal; the consistent advantage is **shorter operative time**.

Salient steps: retroperitoneal laparoscopic nephrectomy

1. **Position:** full flank (90°), table flexed, axillary roll; surgeon behind the patient.
2. **Access (open Hasson):** 1.5–2.5-cm incision just below the 12th rib tip; split muscle, open the lumbodorsal fascia; finger confirms psoas behind and Gerota-covered lower pole in front.
3. **Space creation:** balloon dilator (commercial, or Gaur glove balloon with 600–800 mL saline) anterior to psoas, posterior to Gerota's fascia; blunt-tip balloon-cuff trocar seals the port.
4. **Ports:** two more trocars under vision, one off the 12th rib tip, one posterior and superior to the camera.
5. **Dissection:** along psoas, find the ureter and gonadal vein medially; lift ureter and lower pole to put the hilum on stretch; see arterial pulsation.

6. **Hilum:** clip or staple the **artery first**, then the vein; on the left a **lumbar vein** usually needs division.
7. **Mobilise:** upper pole (adrenal preserved or taken), then lateral and anterior attachments; radical = outside Gerota's fascia.
8. **Extract:** clip ureter, bag the specimen, remove via an extended port; close.

- **Problems:** peritoneal tear (space collapses), emphysema, pneumothorax, vascular injury, flank hernia (intramuscular balloon), rhabdomyolysis (about 1%). **Bail-out:** open the peritoneum and convert.

Diagram to draw: transverse section of the kidney showing the three spaces and Gerota's laminae, with the balloon between psoas and Gerota's fascia.

> **Key points to remember:**
> - Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
> - Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
> - Psoas is the landmark; artery met first.
> - Main advantage: shorter operative time, peritoneum untouched.

Q2. Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child.

Type: LAQ (25 marks) (also answered in part in 2018-Summer Paper 3, Q3a, the treatment algorithm)

Gonadal descent: starting point

- At **7–8 weeks** the gonad lies near the kidney (upper lumbar), held by the **cranial suspensory ligament (CSL)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks, attaches to the **epididymis** and runs to the future inguinal canal.
- **Female:** CSL persists; the gubernaculum becomes the ovarian and round ligaments; the ovary descends only into the pelvis.

Phases of testicular descent (classic two-phase model)

	Transabdominal	Inguinoscrotal
Timing	Weeks 10–15	From the 7th month; inguinal passage rarely before 22 weeks, most testes scrotal after 27 weeks
Events	**CSL regresses**; **gubernaculum swells** (outgrowth); its cord shortens, holding the testis at the internal ring while the kidney and abdomen grow away	Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; bulb then regresses; processus closes
Control	CSL regression by **androgen**; swelling by **INSL3 (insulin-like peptide 3)** from Leydig cells, via its receptor RXFP2	**Androgen**, mainly indirectly via the **genitofemoral nerve (GFN)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure
Failure gives	Intra-abdominal testis	Canalicular or ectopic testis

- **Modern view:** the "transabdominal descent" is largely relative, not true migration. Barteczko and Jacob describe **five phases** (5 weeks to 28 weeks). The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis pass** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster growth.
- **Also know:** AMH (anti-Müllerian hormone) defects (persistent Müllerian duct syndrome) cause cryptorchidism; Insl3-knockout mice are cryptorchid; scrotum forms by 10–12 weeks under DHT (dihydrotestosterone).

Failed step	Result
CSL retained	Intra-abdominal testis
Abnormal or ectopic gubernacular attachment	Canalicular testis; ectopic (superficial inguinal pouch, perineal, femoral, prepenile)
Processus not closed	Hernia, hydrocele
Fibrous remnant of processus	Acquired (ascending) testis

Bilateral undescended testis at 3 years: evaluation

- This is **late presentation**; surgery is advised from 6 months, preferably by 12 months and by 18 months at the latest.

- **History:** prematurity, testes once scrotal (acquired), hernia, family history, salt-wasting illness.
- **Examination:** warm room; supine, then cross-legged or squatting. Sweep from the anterior superior iliac spine down the canal to find the lowest position; check the superficial inguinal pouch. **Retractile** testis stays in the scrotum after gentle traction. Note penis size, meatus (hypospadias), scrotal development, syndromic features.
- **Bilateral palpable:** no investigations needed; plan orchidopexy.
- **Bilateral non-palpable (or with hypospadias or micropenis): treat as DSD (difference of sex development) until proven otherwise.**
  - **Karyotype**; electrolytes and **17-OHP (17-hydroxyprogesterone)** to exclude **CAH (congenital adrenal hyperplasia)** in a virilised 46,XX child.
  - **Hormones:** LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, **AMH and inhibin B**. Measurable AMH or inhibin B confirms testicular tissue; high gonadotrophins with absent AMH suggests **anorchia**. The **hCG (human chorionic gonadotrophin) stimulation test** (testosterone rise) is now less used.
  - Pelvic ultrasound for Müllerian structures.
- **Imaging to localise the testis is not recommended:** ultrasound sensitivity about 45%; it never removes the need for surgery.

Management

- **No hormonal therapy to induce descent:** LHRH (luteinising hormone-releasing hormone) works in about 20%; hCG may increase germ-cell apoptosis. Adjuvant LHRH (buserelin) in bilateral cases is experimental.
- **Palpable:** inguinal orchidopexy (free the cord, ligate the patent processus, retroperitoneal mobilisation, Prentiss manoeuvre if needed, **subdartos pouch**) or scrotal (Bianchi) orchidopexy for low testes. Both sides usually in one sitting.
- **Non-palpable:** examination under anaesthesia, then **diagnostic laparoscopy**:

```
Laparoscopy
 ├─ Testis seen → laparoscopic orchidopexy
 │     vessels short (testis not within 2–4 cm of ring) → Fowler–Stephens
 │     (clip testicular vessels; testis lives on vasal collaterals;
 │      one-stage, or two-stage with second stage at 6 months) or Shehata traction
 ├─ Vessels and vas enter ring → inguinal exploration, excise nubbin
 └─ Blind-ending vessels → stop (vanished testis)
```

- **Both testes very high or of doubtful viability:** stage the sides, the second at 6 months, to avoid bilateral atrophy. Primary orchidopexy (>90% success) beats Fowler–Stephens (60–97%).
- **Biopsy** only if DSD or dysgenesis is suspected. **Bilateral anorchia:** testosterone at puberty, prostheses later.

Follow-up and prognosis

- Check position and size at 6 months.
- **Fertility:** worse when bilateral (oligospermia 42%, azoospermia 31% in one large review).
- **Cancer:** about **3-fold** risk; teach self-examination after puberty.

Diagram to draw: sagittal fetus showing CSL, gubernaculum, processus vaginalis and the two phases with arrows.

Key points to remember:

- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve and CGRP.
- Bilateral non-palpable = DSD work-up first (karyotype, 17-OHP, AMH).
- Laparoscopy decides; no imaging, no hormones.

Q3a. Recist Criteria.

Type: SAQ (10 marks)

Definition and evolution

- **RECIST (Response Evaluation Criteria In Solid Tumours):** standard imaging rules for tumour response and progression.
- **WHO (World Health Organization) 1981** (bidimensional) → **RECIST 2000** (unidimensional) → **RECIST 1.1 (2009)**, now standard. Used in trials for metastatic RCC (renal cell carcinoma), urothelial and prostate cancer.

Lesions at baseline (RECIST 1.1)

Category	Definition
Measurable lesion	Longest diameter **≥10 mm on CT (computed tomography)**, ≥20 mm on chest X-ray, ≥10 mm with callipers
Measurable node	**≥15 mm short axis**; 10–15 mm pathological but non-measurable; <10 mm normal
Non-measurable	Small lesions, blastic bone metastases, effusions, ascites, leptomeningeal disease
Target lesions	Up to **5 in total (maximum 2 per organ)**; largest, reproducible; **sum of diameters** recorded
Non-target	All other lesions, recorded as present or absent

Response categories

Response	Target lesions
CR (complete response)	All disappear; nodes <10 mm
PR (partial response)	**≥30% fall** in sum versus **baseline**
PD (progressive disease)	**≥20% rise** versus **nadir** and **≥5 mm absolute**, or any **new lesion**
SD (stable disease)	Neither PR nor PD

- **Overall response** combines target, non-target and new lesions; confirm response at ≥4 weeks in single-arm trials.
- **End points:** ORR (objective response rate = CR + PR), PFS (progression-free survival), duration of response.

Limitations

- Size only, ignoring necrosis (TKIs (tyrosine kinase inhibitors) shrink tumours little); bone poorly assessed; observer variability.

Modifications

Criteria	Use
iRECIST (2017)	Immunotherapy: first progression is "unconfirmed", confirmed only on the next scan, allowing for **pseudoprogression**
Choi	Density fall counts as response
PCWG3 (Prostate Cancer Working Group 3)	Bone: 12-week scan is the new baseline (flare), then **≥2 new lesions** = progression; PSA (prostate-specific antigen) progression ≥25% and ≥2 ng/mL above nadir; soft tissue by RECIST
PERCIST	PET (positron emission tomography) metabolic response

Diagram to draw: number line of change in sum of diameters: −30% (PR) ... 0 ... +20% (PD).

Key points to remember:

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy pseudoprogression; PCWG3 for prostate.

Q3b. Outline algorithm for difficult catheter placement.

Type: SAQ (10 marks)

Introduction

- **Normal calibre:** meatus 24 Fr (narrowest), bladder neck 28 Fr, prostatic urethra 32 Fr.
- **Causes:** BPH (benign prostatic hyperplasia, commonest), urethral stricture, false passage, meatal stenosis, bladder neck contracture, urethral trauma, abnormal anatomy.
- **Golden rule:** escalate **blind → wire → vision → suprapubic**; never repeat forceful blind passage.

Before starting

- History (surgery, radiotherapy, AUS (artificial urinary sphincter), reconstruction), DRE (digital rectal examination).
- **Absolute contraindication:** suspected urethral injury (pelvic fracture, blood at meatus, high-riding prostate) → **RUG (retrograde urethrogram)**; pelvic fracture urethral injury → suprapubic tube.
- **Technique:** 10–15 mL lubricant (lidocaine gel), penis stretched, steady pressure; inflate only after urine flows at the hub.

Algorithm (Campbell Fig. 15.6)

```
First attempt fails
 ├─ Prostatic obstruction suspected
 │     → 18-Fr coudé, tip (marker) pointing up
 │     → fails → flexible cystoscopy
 └─ Not suspected (stricture likely)
       → smaller catheter
       → fails → hydrophilic glidewire
             → fails → flexible cystoscopy
At cystoscopy / with wire in bladder:
 ├─ Stricture → wire across → 5-Fr catheter, aspirate urine to confirm
 │      → super-stiff wire → serial dilatation (1 Fr above catheter size)
 │      → Councill (open-tip) catheter over wire
 └─ False passage → pass under vision (scope anterior)
        → wire → Councill over wire
Wire cannot be passed / dense stricture / urethral disruption
 → suprapubic catheter (ultrasound-guided percutaneous, or open)
```

- **Why 18-Fr coudé:** stiffer, larger tip area, curved tip follows the bulbar-prostatic angle; 14–16 Fr usually buckle.
- **Blood on the tip** suggests false passage: stop blind attempts.

Special situations

Situation	Approach
AUS	Deactivate; smallest catheter 12–14 Fr; remove early
Obese or female	Exposure: Trendelenburg, retract pannus, two fingers in vagina to guide
Buried penis	Infrapubic pressure; flexible cystoscopy
Neobladder, continent pouch	Never force; scope + wire; suprapubic only with imaging guidance
PUV (posterior urethral valves) neonate	6–8 Fr, or 5-Fr open-ended over glidewire; else suprapubic or vesicostomy

Also know: filiforms, balloon dilators; complications: false passage, haematuria, stricture.

Diagram to draw: coudé catheter following the bulbar-prostatic curve with tip up.

> **Key points to remember:**
>
> - Blind → wire → vision → suprapubic.
> - BPH: 18-Fr coudé, tip up.
> - Councill catheter over wire after dilatation.
> - Suspected urethral injury: RUG first, no blind catheter.

Q3c. Chronic Prostatitis : diagnosis, classification and management.

Type: SAQ (10 marks) (also answered in 2018-Summer Paper 4, Q3d, for CP/CPPS)

Classification: NIH (National Institutes of Health), 1999

Category	Description
I	Acute bacterial prostatitis
II	**Chronic bacterial prostatitis (CBP)**: recurrent UTI (urinary tract infection), **same organism**; well between episodes
IIIA	**CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, inflammatory: leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen
IIIB	CP/CPPS, non-inflammatory (IIIA/IIIB split has no treatment value)
IV	Asymptomatic inflammatory (incidental)

- **CP/CPPS:** pelvic pain **≥3 of the last 6 months**, no uropathogens; 90–95% of cases. Infection, neurogenic inflammation, **pelvic floor dysfunction**, stress and **central sensitisation** interact.

Diagnosis

- **Symptoms:** perineal (commonest), testicular, suprapubic, penile or ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.
- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains.
- **DRE (digital rectal examination):** prostate tender in under half; palpate pelvic floor for **trigger points**.
- **Localisation tests:**
  - **Meares–Stamey 4-glass:** VB1 (urethra), VB2 (bladder), EPS, VB3 (prostate).
  - **2-glass test** (pre- and post-massage urine): predicts the 4-glass result in >95%.
- Urinalysis, culture, PVR (post-void residual), flow; cytology if irritative symptoms; cystoscopy if low flow or bladder pain. PSA (prostate-specific antigen) not indicated.
- **UPOINT phenotyping:** Urinary, Psychosocial, Organ-specific, Infection, Neurological/systemic, Tenderness of muscles.

Management

Category II (CBP): fluoroquinolone (lipid-soluble zwitterion, enters prostate) for **4–6 weeks** (EAU (European Association of Urology)); Chlamydia: azithromycin; resistant: TMP-SMX (trimethoprim-sulfamethoxazole) 3 months; refractory: suppression or TURP (transurethral resection of prostate).

Category III: multimodal, UPOINT-directed (monotherapy fails)

Domain	Therapy
Urinary	α-blocker if recent onset and α-blocker naive
Psychosocial	CBT (cognitive behavioural therapy), stress reduction
Organ	5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia); quercetin
Infection	Quinolone ≥6 weeks only if treatment-naive and <1 year; no repeated courses
Neurological	Pregabalin, amitriptyline, duloxetine
Tenderness	**Pelvic floor physiotherapy, myofascial release**, biofeedback

- **Others:** acupuncture (responders 61.5% vs 38.3% sham), aerobic exercise, PTNS (percutaneous tibial nerve stimulation). **Not recommended:** TURP or prostatectomy for pain.

Diagram to draw: UPOINT snowflake with one therapy per domain.

> **Key points to remember:**
> - NIH I–IV; III = pain ≥3 of 6 months, no infection.
> - CBP: same organism, fluoroquinolone 4–6 weeks.
> - CP/CPPS: NIH-CPSI to score, UPOINT to treat, multimodal.

Q3d. Multiparametric MRI : its components and interpretation for prostate cancer.

Type: SAQ (10 marks)

Introduction

- **mpMRI (multiparametric magnetic resonance imaging):** anatomical plus functional sequences, usually at 3 T without endorectal coil. Reported by **PI-RADS v2.1 (Prostate Imaging Reporting and Data System, 2019)**.

Components

Sequence	Principle	Cancer appearance
T2W (T2-weighted), triplanar	Anatomy; normal PZ (peripheral zone) bright	Low-signal lesion; "erased charcoal" in TZ (transition zone); shows EPE (extraprostatic extension) and SVI (seminal vesicle invasion)
DWI (diffusion-weighted imaging) with **ADC (apparent diffusion coefficient)** map; high b-value (≥1400)	Cellular tumour restricts water movement	Bright on high-b DWI, **dark on ADC**; ADC falls as Gleason rises
DCE (dynamic contrast-enhanced) T1 with gadolinium	Angiogenesis: early enhancement and washout	Early focal enhancement (type 3 curve)

- **Biparametric MRI (T2W + DWI):** no contrast, cheaper, similar detection by meta-analysis; poorer staging.
- **PI-QUAL (Prostate Imaging Quality):** quality score (1–5) before reporting.

Interpretation: PI-RADS v2.1

Score	Clinically significant cancer
1	Highly unlikely
2	Unlikely
3	Equivocal
4	Likely; lesion <1.5 cm
5	Highly likely; ≥1.5 cm or definite EPE

Zone	Dominant sequence	Tie-breaker
Peripheral	**DWI**	DWI 3 + DCE positive → 4
Transition	**T2W**	T2W 3 + DWI 5 → 4; T2W 2 + DWI ≥4 → 3

- **DCE only upgrades PZ 3 to 4**; it never changes 4–5 or TZ scores.
- Cancer in about one-third of PI-RADS 3, over half of 4 and about 90% of 5.
- **PRECISE** score (1–5) tracks change during active surveillance.

Clinical use and evidence

- **PROMIS (2017):** sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI pathway with targeted biopsy of PI-RADS ≥3 detected more significant cancer (38% vs 26%), 28% avoided biopsy.

- EAU (European Association of Urology) and AUA (American Urological Association): MRI **before biopsy**; also staging, active surveillance, focal therapy planning.
- Limitations: low specificity (Cochrane 0.37); reader-dependent; misses some significant cancers.

Diagram to draw: prostate zones (PZ, TZ) with a PZ lesion dark on ADC and T2.

> **Key points to remember:**
> - T2W anatomy, DWI/ADC cellularity, DCE vascularity.
> - PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
> - PRECISION and PROMIS: MRI before biopsy.

Q3e. Indications for outpatient cystourethroscopy.

Type: SAQ (10 marks)

Introduction

- **Flexible cystourethroscopy** under local anaesthesia is the **commonest office procedure** in urology; no lithotomy needed.
- **Prerequisites:** consent; treat any UTI (urinary tract infection) first; lidocaine gel with **≥5 minutes dwell** (ideally 10–20); no antibiotic prophylaxis unless risk factors (age, diabetes, immunosuppression, catheters, prior UTI).

Diagnostic indications

Indication	Comment
Haematuria	All **gross haematuria** (with CT (computed tomography) urography, cytology). Microhaematuria by AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) risk: intermediate and high risk; shared decision in low risk
Bladder cancer surveillance	NMIBC (non-muscle-invasive bladder cancer): first check at 3 months; low risk then yearly; high risk 3–4-monthly for 2 years; also upper-tract follow-up of UTUC (upper tract urothelial carcinoma)
Recurrent UTI	Look for stone, diverticulum, stricture, fistula, foreign body
LUTS (lower urinary tract symptoms)	Not routine; before invasive BPH (benign prostatic hyperplasia) treatment (median lobe before urethral lift, bladder-neck contracture, stricture)
Pain syndromes	IC/BPS (interstitial cystitis/bladder pain syndrome): **Hunner lesion**; chronic pelvic pain with bladder pain or low flow
Women	Suspected urethral diverticulum, vesicovaginal fistula, mesh or suture erosion after sling, recurrent UTI after surgery
Others	Urethral stricture assessment and follow-up; abnormal urine cytology; radiation cystitis; foreign body

Therapeutic indications

- **Removal or exchange of ureteric stents** (exchange with fluoroscopy).
- **Biopsy and fulguration** of small (≤1 cm) low-grade recurrences; office laser ablation.
- **Intradetrusor botulinum toxin** injection (OAB (overactive bladder), neurogenic detrusor overactivity).
- Urethral bulking agents; stricture dilatation; small stones or foreign bodies; blue-light cystoscopy where available.

When not indicated or done in theatre

- Not needed for the index stress incontinence patient, initial OAB or uncomplicated LUTS.
- **No surveillance cystoscopy** in chronic indwelling catheters (AUA/SUFU 2021).
- Theatre if TURBT (transurethral resection of bladder tumour) needed, poor tolerance, children, or complex diversions.

Diagram to draw: flowchart of haematuria risk group leading to cystoscopy plus imaging.

Key points to remember:

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, Botox, fulguration of small recurrences.

Q3f. Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each.

Type: SAQ (10 marks) (mini-PCNL: see 2022-Summer Paper 1, Q1)

Indications for PCNL (percutaneous nephrolithotomy)

- Renal stones **>2 cm**, staghorn, lower pole >1 cm, calyceal diverticulum, failed retrograde route or urinary diversion, UPJ (ureteropelvic junction) obstruction.

Comparison

Feature	Prone (Goodwin, 1955)	Supine (Valdivia-Uría, 1987)
Positioning	Lithotomy for ureteric catheter, then turned prone; chest and pelvic bolsters (horizontal bring kidney down more); arms ≤90°, face and knees padded	Ipsilateral flank at table edge on a bolster, arm across chest; **Galdakao-modified (Scoffone)**: legs in lithotomy for simultaneous retrograde access
Best suited	Standard anatomy; **upper pole and posterior calyx** access; multiple tracts	Obese, cardiopulmonary disease, kyphosis; ECIRS (endoscopic combined intrarenal surgery); ileal conduit or continent channel
Technique	Posterior calyx puncture under fluoroscopy or ultrasound; tract more vertical; large working area	Puncture in posterior axillary line; tract more horizontal and longer; kidney more mobile; air outlines anterior calyces
Drainage	Against gravity; higher pelvic pressure	**Gravity drains fragments**, low intrarenal pressure
Anaesthesia	Airway access difficult; turning needed	**Airway retained**; no repositioning

Specific complications

- **Prone:** ocular injury, neuropraxia (brachial plexus), pressure injury, airway tube dislodgement while turning.
- **Supine: colon injury** (if puncture too anterior), access failure to upper pole, kidney displacement during dilatation.
- **Both:** bleeding (transfusion now about 1%), pleural injury above the 12th rib, sepsis, bowel, spleen or liver injury.

Advantages and disadvantages

	Prone	Supine
Advantages	Wide working area, easy upper pole access, familiar	Single position, shorter time, ECIRS, gravity clearance, safer anaesthesia
Disadvantages	Repositioning, positioning injuries, needs split-leg table for ECIRS	Smaller working area, collapsed system, longer tract, harder upper pole

- **Evidence:** stone-free rates and complications similar; supine often shortens operating time; choice is mainly surgeon preference.

Diagram to draw: cross-section showing tract direction prone (vertical, posterior calyx) versus supine (horizontal, posterior axillary line).

> **Key points to remember:**
> - Prone: posterior and upper pole access, big working space.
> - Supine: airway, ECIRS, gravity drainage, no turning.
> - Similar outcomes; choose by anatomy and experience.

Rapid Revision Sheet – 2024-Winter Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. (LAQ)

- Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
- Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
- Psoas is the landmark; artery met first.
- Main advantage: shorter operative time, peritoneum untouched.

Q2. Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child. (LAQ)

- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve and CGRP.
- Bilateral non-palpable = DSD work-up first (karyotype, 17-OHP, AMH).
- Laparoscopy decides; no imaging, no hormones.

Q3a. Recist Criteria. (SAQ)

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy pseudoprogression; PCWG3 for prostate.

Q3b. Outline algorithm for difficult catheter placement. (SAQ)

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

Q3c. Chronic Prostatitis : diagnosis, classification and management. (SAQ)

- NIH I–IV; III = pain ≥3 of 6 months, no infection.
- CBP: same organism, fluoroquinolone 4–6 weeks.
- CP/CPPS: NIH-CPSI to score, UPOINT to treat, multimodal.

Q3d. Multiparametric MRI : its components and interpretation for prostate cancer. (SAQ)

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.

Q3e. Indications for outpatient cystourethroscopy. (SAQ)

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, Botox, fulguration of small recurrences.

Q3f. Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each. (SAQ)

- Prone: posterior and upper pole access, big working space.
- Supine: airway, ECIRS, gravity drainage, no turning.
- Similar outcomes; choose by anatomy and experience.